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REVIEW

# BRCA1 and BRCA2: from cancer susceptibility to synthetic lethality

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**The discovery of *BRCA1* and *BRCA2* as tumor susceptibility genes and their role in genome maintenance has transformed our understanding of hereditary breast and ovarian cancer. This review traces the evolution of BRCA1/2 research over the past 30 years, highlighting key discoveries in the field and their contributions to tumor development. Additionally, we discuss current preventive measures for *BRCA1/2* mutation carriers and targeted treatment options based on the concept of synthetic lethality. Finally, we explore the challenges of acquired therapy resistance and discuss potential alternative avenues for targeting *BRCA1/2* mutant tumors.**

Supplemental material is available for this article.

In the mid-1990s, mutations in the high-risk cancer susceptibility genes *BRCA1* and *BRCA2* were first linked to the development of hereditary breast and ovarian cancer (HBOC) (Miki et al. 1994; Wooster et al. 1995). Currently, it is believed that a vast majority of HBOCs are caused by deleterious heterozygous germline mutations in *BRCA1/2*, which significantly elevate the risk of developing breast, ovarian, prostate, and pancreatic cancer (Petrucelli et al. 1993; Waddell et al. 2015; Pritchard et al. 2016; Yoshida 2021). Despite their similar cellular functions, *BRCA1* and *BRCA2* mutated tumors exhibit distinct molecular and clinical features. *BRCA1* mutation carriers mainly develop triple-negative breast cancer (TNBC), characterized by early onset and a high risk of relapse, and *BRCA2* mutation carriers mostly develop luminal-like breast cancers that are generally responsive to adjuvant therapy with antiestrogens (Mavaddat et al. 2012; Loibl et al. 2024).

In this review, we outline the key discoveries since the identification of *BRCA1/2* nearly 30 years ago (Fig. 1). Additionally, we summarize the biological and molecular functions of BRCA1/2 and explore current and potential future therapeutic avenues for *BRCA1/2* mutated tumors.

## Discovery of BRCA1 and BRCA2 and early functional insights

More than a century after Broca (1866) reported on the existence of hereditary breast cancer, a locus on chromosome 17q21 was found to confer a marked increase in breast cancer risk (Hall et al. 1990; Lenoir 1991). Four years later, Miki et al. (1994) identified the *BRCA1* gene and detected mutations in its coding sequence in breast cancer patients. Subsequently, *BRCA2* was discovered on 13q12.3 by studying breast cancer cases without *BRCA1* linkage (Wooster et al. 1995). In 2006, the third high-risk hereditary breast cancer gene, *PALB2*, was identified as a partner of BRCA2 (Xia et al. 2006; Erkko et al. 2007; Rahman et al. 2007). Since their initial discoveries, several functional domains have been identified within BRCA1/2 and PALB2, hinting at their cellular functions (Fig. 2).

Although early reports proposed that BRCA1 may function as a transcription factor (Chapman and Verma 1996; Monteiro et al. 1996), additional research revealed that BRCA1 and RAD51 colocalize in nuclear foci, suggesting a role in genome stability (Scully et al. 1997b). Initial functional experiments showed that mouse embryonic fibroblasts isolated from *Brca1* or *Brca2* knockout embryos spontaneously develop abnormal chromosomal structures during early tissue culture passages (Shen et al. 1998; Tutt et al. 1999). In line with this, repair of DNA double-strand breaks (DSBs) via homologous recombination (HR) was found to be impaired in *BRCA1* mutated cell lines, confirming a role for BRCA1 in the maintenance of genome stability via HR (Moynahan et al. 1999; Scully et al. 1999). Additionally, BRCA1 and BRCA2 have been implicated in both the activation and the maintenance of the G2/M cell cycle checkpoint to prevent segregation of damaged chromosomes into daughter cells (Xu et al. 2001a; Yarden et al. 2002; Menzel et al. 2011; Shabbeer et al. 2013; Simhadri et al. 2019). Yeast and

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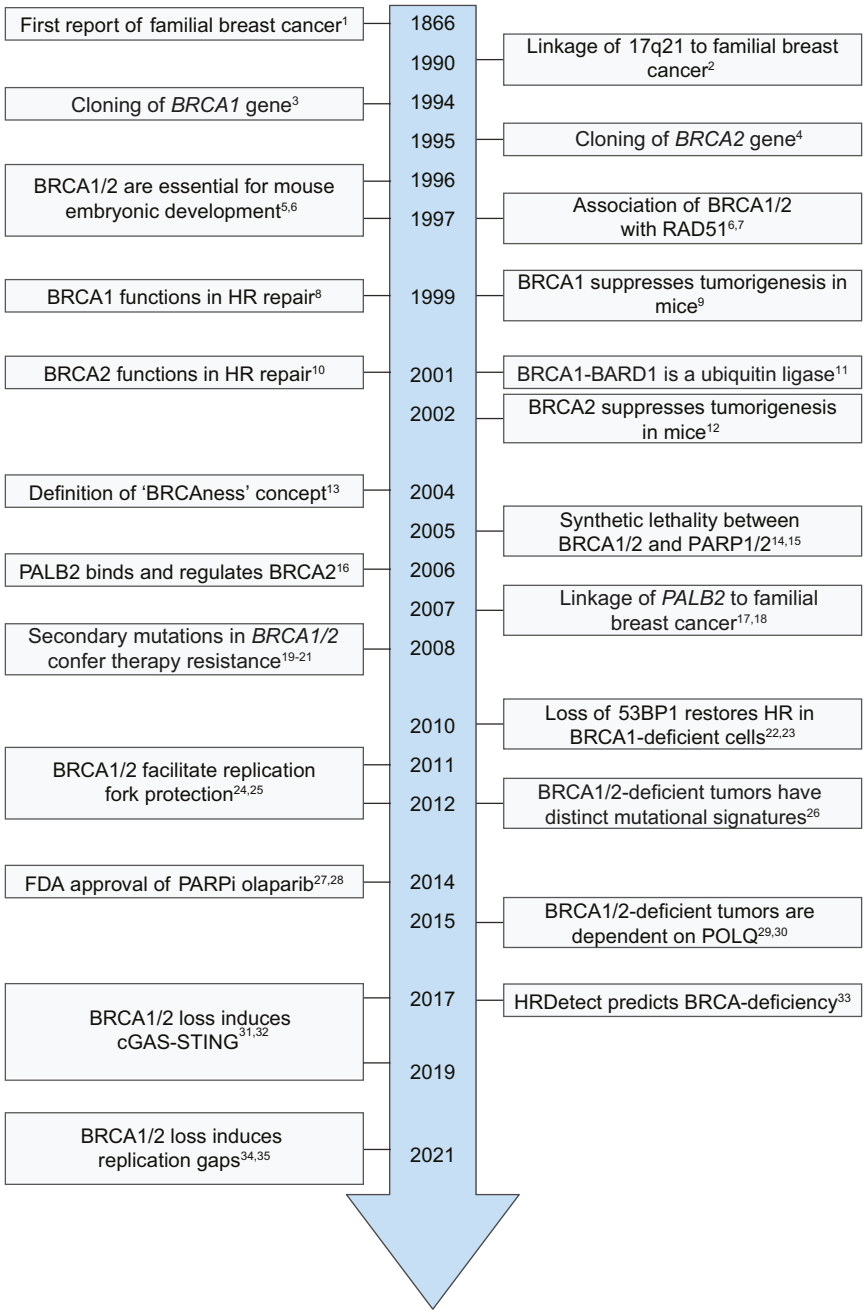



**Figure 1.** Chronology of the major discoveries in BRCA1 and BRCA2 research (numbers 1–35 refer to the superscript numbers in the figure): (1) Broca (1866), (2) Hall et al. (1990), (3) Miki et al. (1994), (4) Wooster et al. (1995), (5) Hakem et al. (1996), (6) Sharan et al. (1997), (7) Scully et al. (1997b), (8) Moynahan et al. (1999), (9) Xu et al. (1999), (10) Moynahan et al. (2001a), (11) Hashizume et al. (2001), (12) Jonkers et al. (2001), (13) Turner et al. (2004b), (14) Farmer et al. (2005), (15) Bryant et al. (2005), (16) Xia et al. (2006), (17) Erkko et al. (2007), (18) Rahman et al. (2007), (19) Sakai et al. (2008), (20) Edwards et al. (2008), (21) Swisher et al. (2008), (22) Bunting et al. (2010), (23) Bouwman et al. (2010), (24) Schlacher et al. (2011), (25) Schlacher et al. (2012), (26) Nik-Zainal et al. (2012), (27) Ledermann et al. (2012), (28) Kaufman et al. (2015), (29) Mateos-Gomez et al. (2015), (30) Ceccaldi et al. (2015), (31) Parkes et al. (2017), (32) Heijink et al. (2019), (33) Davies et al. (2017), (34) Tirman et al. (2021), and (35) Cong et al. (2021).

mammalian two-hybrid analyses showed that BRCA1 interacts with BARD1, and further research suggested that the BRCA1–BARD1 heterodimer possesses E3 ubiquitin ligase activity (Wu et al. 1996; Meza et al. 1999; Hashizume et al. 2001).

Initial studies on BRCA2 indicated that it interacts with RAD51 and is essential for cellular proliferation and response to genotoxic agents, hinting at a function in DNA repair and genome stability (Sharan et al. 1997; Patel et al. 1998). Cells lacking the BRCA2 C terminus, which is essential for its interaction with RAD51, exhibit increased sensitivity to γ-irradiation but maintain effective G1–S and G2–M checkpoints after exposure, suggesting that loss of BRCA2 confers genomic instability due to defective RAD51-mediated DNA repair rather than defective cell cycle regulation (Morimatsu et al. 1998). Additional studies revealed that BRCA2 regulates the intracellular localization and DNA binding ability of RAD51 and that BRCA2 loss reduces HR activity upon the induction of DSBs (Davies et al. 2001; Moynahan et al. 2001b).

These early findings established crucial roles for BRCA1 and BRCA2 in HR repair, providing a basis for subsequent research that revolutionized our understanding of DNA repair and its association with tumorigenesis.

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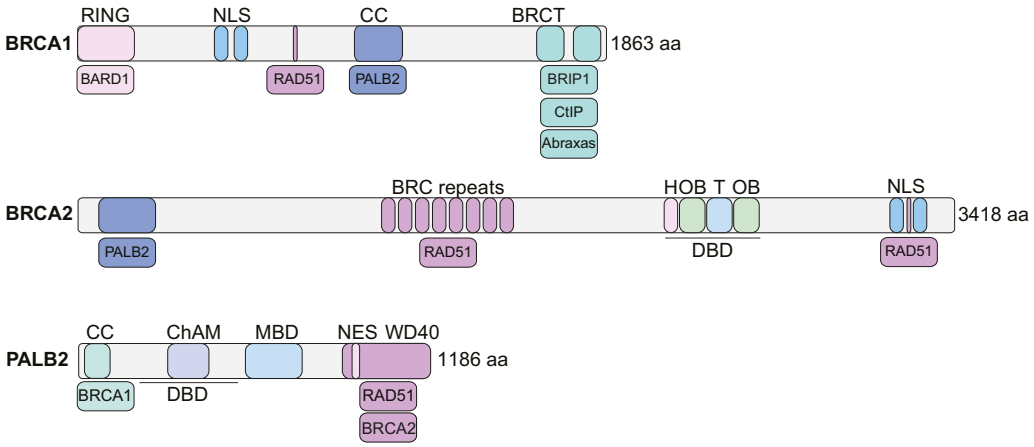



**Figure 2.** BRCA1/2 and PALB2 functional domains. Schematic representation of the functional domains of BRCA1, BRCA2, and PALB2. (*Top*) BRCA1 contains an N-terminal RING domain that possesses E3 ubiquitin ligase activity and interacts with BARD1; a nuclear localization signal (NLS) domain; a coiled-coil (CC) domain that associates with PALB2; and a C-terminal BRCT domain interacting with BRIP1, Abraxas, and CtIP. (*Middle*) BRCA2 contains a PALB2-interacting domain; eight conserved BRC repeats that are crucial for RAD51 binding; a DNA binding domain (DBD) comprised of a helical domain (H), three oligonucleotide-binding (OB) folds, and a tower domain (T); an NLS harboring cyclin-dependent kinase (CDK) phosphorylation sites; and a C-terminal domain important for binding of RAD51 during the replication stress response. (*Bottom*) PALB2 contains an N-terminal CC domain binding BRCA1, a chromatin-associated motif (ChAM) lying within the DBD, an MRG-15 binding domain (MBD), and C-terminal WD40 repeats containing a nuclear export signal (NES).

## Roles of BRCA1/2 in the DNA damage response

As described above, BRCA1 and BRCA2 play crucial roles in HR and the maintenance of genome stability. In recent years, numerous studies have demonstrated that BRCA1/2 also facilitate the repair of other DNA lesions (Fig. 3), which is discussed below.

### *Homologous recombination repair*

Early observations showed that BRCA1 and BRCA2 colocalize with RAD51 foci, and a direct role for their involvement in HR was demonstrated using a reporter construct soon after (Scully et al. 1997b; Sharan et al. 1997; Moynahan et al. 1999, 2001b).

Upon occurrence of a DSB and initiation of the DNA damage response (DDR), BRCA1–BARD1 is recruited and ubiquitinates histone H2A to trigger the eviction of 53BP1, a factor that prevents HR initiation via DNA end resection and thereby promotes classical nonhomologous end joining (cNHEJ) (Bunting et al. 2010; Densham et al. 2016). In line with the importance of BRCA1–BARD1 for HR, BARD1 ablation phenocopies BRCA1 loss, resulting in decreased RAD51 focus formation and impaired HR (McCarthy et al. 2003; Zhao et al. 2017). Interestingly, BRCA1 appears dispensable for the initiatory steps of HR in the absence of its competitor, 53BP1, as the loss of 53BP1 partially restores HR and improves survival of BRCA1-deficient cells (Cao et al. 2009; Bouwman et al. 2010; Bunting et al. 2010).

Downstream, BRCA1 interacts with the endonuclease CtIP through its BRCT domain and colocalizes with members of the MRN (MRE11–RAD50–NBS1) complex to catalyze DNA end resection (Figs. 2, 3; Chen et al. 2008). Once single-stranded DNA (ssDNA) overhangs are generated, they are bound by the ssDNA-binding replication protein A (RPA). Subsequently, RPA is replaced by RAD51, facilitated by BRCA1–PALB2–BRCA2, to promote invasion of the resected 3′ ssDNA end into the homologous repair template and to aid homology search (Sy et al. 2009; Zhang et al. 2009a; Liu et al. 2010; Thorslund et al. 2010). Specifically, BRCA2 enables RPA replacement by RAD51 and stabilizes RAD51 filaments by inhibiting ATP hydrolysis (Jensen et al. 2010).

In addition to its role in HR-dependent repair of canonical double-ended DSBs, BRCA1 has been reported to act at single-ended DSBs arising at stalled replication forks by promoting HR and preventing the accumulation of tandem duplications (Willis et al. 2014, 2017). Interestingly, BRCA1 is dispensable for promoting DSB end resection at collapsed forks but is still required for subsequent RAD51 loading, indicating that the role of BRCA1 in HR repair of DSBs depends on the context in which they are generated (Pavani et al. 2024).

To maintain genome stability, BRCA1 forms distinct complexes with various DNA repair factors. Structurally, BRCA1 and PALB2 both contain coiled-coil (CC) domains that directly interact with each other, facilitating the formation of a large BRCA1–PALB2–BRCA2–RAD51 (BRCA1-P) complex that promotes RAD51 recruitment and filament formation at sites of DNA damage (Sy et al. 2009; Zhang et al. 2009a,b). The localization of BRCA1-P to DSBs is regulated by RNF168-mediated ubiquitination of H2A, which triggers the recruitment of BARD1. This in turn leads to BARD1–BRCA1-mediated accumulation of PALB2–RAD51 at DNA breaks (Krais et al. 2021). The BRCA1-A complex, which consists of RAP80, BRCC36, BRCC45, MERIT40, and Abraxas, aids

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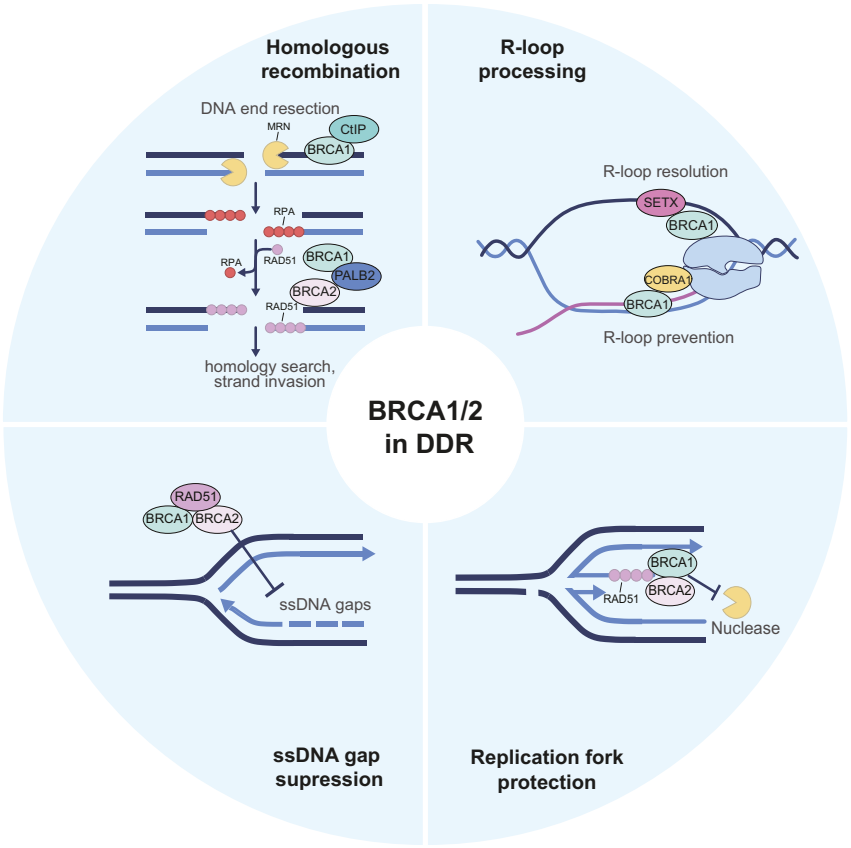



**Figure 3.** Functions of BRCA1/2 in the DDR. BRCA1/2 are involved in several genome maintenance processes, including faithful DNA repair via HR (*top left*), ssDNA gap suppression (*bottom left*), replication fork protection (*bottom right*), and prevention and resolution of R-loops (*top right*).

in recruiting BRCA1 to DSBs by binding to RNF8-induced K63-linked polyubiquitin chains on histone H2AX (Kim et al. 2007a,b; Liu et al. 2007b; Sobhian et al. 2007; Wang and Elledge 2007). Additionally, this complex has been shown to ensure accurate activation of the G2/M checkpoint upon occurrence of DNA damage (Kim et al. 2007a; Liu et al. 2007b; Sobhian et al. 2007; Wang et al. 2007; Coleman and Greenberg 2011; Hu et al. 2011) and prevent hyperresection of DSBs by modulating the binding of BRCA1 to other proteins (Fig. 2; Hu et al. 2011). The BRCA1-B complex, composed of TOPBP1 and BRIP1/FANCJ/BACH1 (Yu et al. 2003; Greenberg et al. 2006), is thought to assemble during S phase and mediate replication stress-induced checkpoint activation and interstrand cross-link repair (ICL) (Litman et al. 2005; Gong et al. 2010). Last, BRCA1, CtIP, and MRN form the BRCA1-C complex, which is essential for determining the speed of DNA end resection during early steps of HR (Chen et al. 2008; Cruz-García et al. 2014).

A systematic analysis of the BRCA1-A, BRCA1-B, and BRCA1-C complexes revealed that loss of the BRCA1-B component BRIP1 or the BRCA1-C component CtIP, but not the BRCA1-A component RAP80, results in reduced cell proliferation and increased sensitivity to ICL-inducing agents (Li et al. 2024). Although both RAP80 and CtIP are involved in the recruitment of BRCA1 to sites of DNA damage, RAD51 focus formation is strongly reduced upon loss of CtIP but only mildly affected by loss of RAP80 (Li et al. 2024). This suggests that BRCA1 recruitment is mediated by several routes. Further research is required to better understand the direct functions of these different BRCA1 complexes in genome stability and their possible involvement in other BRCA1-related pathways.

### *Interstrand cross-link repair*

Besides their function in HR, BRCA1/2 are also important for repairing ICL lesions (Garcia-Higuera et al. 2001; Howlett et al. 2002). ICLs are complex DNA lesions that covalently link the two strands of the double helix, thereby obstructing replication and transcription (Deans and West 2011). Mutations in ICL repair genes cause Fanconi anemia (FA), a syndrome associated with cancer predisposition, congenital defects, and bone marrow failure (Rosenberg et al. 2003). Mutations in *BRCA1*/*FANCS*, *BRCA2*/*FANCD1*, and *PALB2*/*FANCN* have been identified in FA patients (Howlett et al. 2002; Reid et al. 2007; Xia et al. 2007; Sawyer et al. 2015). Similar to canonical FA pathway members, loss of BRCA1/2 causes sensitivity to ICL-inducing agents such as mitomycin C and cisplatin, leading to genomic instability (Bhattacharyya et al. 2000; Yu et al. 2000; D'Andrea and Grompe 2003).

Once replication forks stall at an ICL, the lesion is recognized by the FA core complex and further processed through nucleolytic activity, resulting in the generation

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of a DSB (Deans and West 2011). Although the loss of 53BP1 rescues HR in BRCA1-deficient cells, these cells remain hypersensitive to cross-linking agents, providing further evidence that BRCA1 has an HR-independent role in ICL repair (Bunting et al. 2012). Further downstream in ICL repair, the BRCA1–PALB2–BRCA2 complex mediates the formation of RAD51 nucleofilaments on the generated DSB, similar to its function in HR (Deans and West 2011; Hashimoto et al. 2016).

### *R-loop processing*

More recent studies have shown that BRCA1/2 also prevent the accumulation of RNA:DNA hybrids, termed R-loops (Fig. 3; Hatchi et al. 2015; Shivji et al. 2018). R-loops are three-stranded nucleic acid structures formed during transcription when RNA reanneals with its DNA template behind the advancing RNA polymerase. Although these structures are important for various cellular processes, they also pose a threat to genome stability if not resolved in a timely fashion (Brickner et al. 2022).

BRCA1 not only physically interacts with Senataxin (SETX), the major RNA helicase required for R-loop dissolution, but also forms a complex with SETX to be recruited to highly transcribed regions (Hill et al. 2014; Hatchi et al. 2015). BRCA1 may also prevent the accumulation of Pol II pausing-induced R-loops through its interaction with COBRA1/NELFB (Ye et al. 2001). Indeed, ablation of COBRA1 reduced R-loop accumulation and suppressed tumorigenesis in *Brca1* mutant mammary epithelium, demonstrating functional antagonism between BRCA1 and COBRA1 (Zhang et al. 2017). However, it remains unclear whether R-loop accumulation contributes to *Brca1*-associated tumorigenesis, because a recent study showed that R-loop removal by RNase H1 overexpression in *Brca1* mutant mouse mammary epithelia does not reduce tumor incidence (Chiang et al. 2024). Collectively, these studies show that BRCA1 has a dual role in preventing R-loop accumulation: On the one hand, it attenuates R-loops induced by the Pol II-pausing protein COBRA1, and on the other hand, it resolves R-loops by recruiting SETX.

BRCA2 has also been implicated in R-loop biology, as BRCA2-depleted cells display increased R-loop levels. BRCA2 may be recruited by the mRNA export complex TREX-2 to promote R-loop dissolution and replication fork restart (Bhatia et al. 2014). Additionally, BRCA2 may be recruited by the zinc finger protein ZFP281 to bivalent chromatin regions to promote replication through persistent R-loops (Wang et al. 2022). Furthermore, BRCA2 directly interacts with RNA Pol II at promoter-proximal pausing sites to ensure controlled RNA Pol II release, thereby preventing unscheduled R-loop generation (Shivji et al. 2018).

In summary, both BRCA1 and BRCA2 are implicated in R-loop processing to maintain genome stability. However, it remains unclear whether BRCA1/2 are directly involved in R-loop clearance or whether they are recruited by and interact with accessory factors such as RNA Pol II. Although R-loops are a well-established source of genome instability, their specific contribution to BRCA1/2-associated tumorigenesis remains to be determined.

## Roles of BRCA1/2 at replication forks

Early observations reported that BRCA1 and BRCA2 colocalize with PCNA upon replicative stress (Chen et al. 1998). Only much later was it discovered that both proteins are essential for protecting stalled DNA replication forks and suppressing postreplicative ssDNA gaps (Fig. 3; Schlacher et al. 2011, 2012; Cong et al. 2021; Panzarino et al. 2021).

### *Replication fork protection*

While the exact mechanisms remain to be elucidated, current evidence suggests that BRCA1 and BRCA2 stabilize RAD51 at stalled or reversed forks. Although both HR and fork protection require RAD51 loading and stabilization, BRCA1 and BRCA2 have distinct functions in these processes. For example, the BRCA1–PALB2 interaction is important for HR but dispensable for fork protection. Instead, the proline isomerase PIN1 promotes the interaction of the BRCA1–BARD1 complex with RAD51, facilitating RAD51 recruitment to stalled forks (Daza-Martin et al. 2019). In BRCA2, a C-terminal RAD51-binding site is critical for fork protection but dispensable for HR and resistance to PARP inhibitors (Schlacher et al. 2011).

In the absence of BRCA1/2, inefficient RAD51 loading and stabilization cause excessive fork degradation by the MRE11 nuclease (Schlacher et al. 2011, 2012). This fork degradation process has been extensively studied, revealing multiple factors that regulate MRE11 recruitment to stalled forks in the absence of BRCA1/2, such as PTIP, PARP1 (Ding et al. 2016; Ray Chaudhuri et al. 2016), and RAD52 (Jensen et al. 2010; Liu et al. 2010; Thorslund et al. 2010; Kolinjivadi et al. 2017; Mijic et al. 2017).

In vitro studies also confirm that BRCA2 is loaded onto chromatin during replication and stimulates RAD51-mediated strand exchange at ssDNA/dsDNA junctions (Jensen et al. 2010; Liu et al. 2010; Thorslund et al. 2010). This RAD51 filament formation is thought to recruit Polα, which then displaces RAD51 to fill in ssDNA gaps (Kolinjivadi et al. 2017). Conversely, BRCA2 seems to be dispensable for RAD51-mediated fork reversal but is required for RAD51 stabilization during fork stalling. (Zellweger et al. 2015; Ray Chaudhuri et al. 2016; Mijic et al. 2017; Lim et al. 2024).

Although both BRCA1 and BRCA2 protect replication forks, they do so via different mechanisms. For instance, loss of the nucleosome remodeling factor CHD4 and the structure-specific endonuclease MUS81 rescues fork protection in BRCA2-deficient cells but not in BRCA1-deficient cells (Guillemette et al. 2015; Ray Chaudhuri et al. 2016; Rondinelli et al. 2017). These differences suggest that the loss of either protein may activate different alternative fork processing pathways or generate different replication intermediates.

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*ssDNA gap suppression*

Postreplicative ssDNA gaps typically form on both the leading and lagging strands, as well as upon replication fork stalling in response to genotoxic stress. Replication forks that arrest upon nucleotide depletion or genotoxic stress may reverse to bypass the lesions or to facilitate replication-coupled repair. Alternatively, PRIMPOL-mediated repriming can occur downstream from the lesion, but this process results in the formation of ssDNA gaps (Mourón et al. 2013). Postreplicative ssDNA gaps also arise during Okazaki fragment processing (OFP) on the lagging strand. These gaps are thought to be recognized by PARP1 (Hanzlikova et al. 2018) and are sealed by the flap endonucleases FEN1 and LIG1, along with possibly other nucleases (Sun et al. 2023).

Current evidence indicates that BRCA1 and BRCA2 are essential for both replication fork stalling and repriming processes. BRCA2 inhibits PRIMPOL-mediated repriming and prevents MRE11-dependent fork degradation (Kolinjivadi et al. 2017; Tirman et al. 2021). Interestingly, in BRCA1/2-deficient cells, the resulting ssDNA gaps coincide with elevated S-phase PARylation and are resistant to MRE11 degradation, suggesting that these gaps originate from defective OFP rather than stalled forks (Cong et al. 2021; Paes Dias et al. 2021). In support of this hypothesis, the loss of FEN1 is synthetic lethal with BRCA1/2 deficiency (Mengwasser et al. 2019). However, it is important to note that FEN1 is also essential for functional base excision repair (BER) and microhomology-mediated end joining (MMEJ); therefore, it remains unclear which of these roles are important for the survival of BRCA1/2-deficient cells.

To summarize, it is becoming clear that BRCA1 and BRCA2 have distinct roles in HR and replication. However, it remains to be determined to what extent the loss of fork protection and gap suppression contributes to genome instability and cancer predisposition induced by BRCA1/2 loss. *BRCA1/2* heterozygous cells display normal HR levels but impaired fork protection and suppression of ssDNA gaps, which might foster the accumulation of tumor-promoting mutations, including loss of the wild-type *BRCA1/2* allele (Pathania et al. 2014; Lim et al. 2024). Mutations in BARD1 disrupting BRCA1/BARD1-mediated fork protection but not affecting HR are not tumorigenic in mice (Billing et al. 2018). Similarly, analysis of separation-of-function mutants of *BRCA2* showed that loss of fork protection and gap suppression does not induce genome instability or tumorigenesis (Lim et al. 2024). This suggests that HR may be the ultimate driver of tumorigenesis; however, further research is needed to address this question.

## Roles of BRCA1/2 in normal development and tumor suppression

*Insights from genetically engineered mouse models*

Mouse models have proven highly valuable for exploring various aspects of BRCA1/2 biology, offering key insights into their roles in normal development and tumor suppression and therapy response of *BRCA1/2* mutant tumors (Supplemental Table S1).

Soon after the discovery of BRCA1, reports demonstrated that complete loss of BRCA1 in mice results in early embryonic lethality, indicating its crucial role in maintaining high proliferation rates in early embryos (Hakem et al. 1996; Ludwig et al. 1997). Several studies generated *Brca1* mutant mice by targeting the largest exon of *Brca1*, exon 11. Although these mouse models express an alternatively spliced isoform of *Brca1* lacking exon 11, loss of full-length *Brca1* still conferred early embryonic lethality (Gowen et al. 1996; Liu et al. 1996; Shen et al. 1998). Similarly, complete loss of BRCA2 also causes early embryonic lethality in mice, phenocopying the loss of RAD51 and BRCA1 (Ludwig et al. 1997; Sharan et al. 1997). These findings underscore an essential role for BRCA1/2 in maintaining cellular proliferation during early developmental stages. Although homozygous *Brca2* mutant mice with a C-terminal truncation mutation survived to adulthood and developed thymic lymphomas (Connor et al. 1997; Friedman et al. 1998), heterozygous *Brca1/2* knockout animals did not exhibit significant tumor predisposition, suggesting that additional (epi)genetic alterations are required.

Therefore, models with additional permutations were generated (Supplemental Table S1). Whereas homozygous loss of *Trp53* only marginally extended the survival of homozygous *Brca1*-null embryos, p53 dosage reduction via heterozygous *Trp53* loss was sufficient to rescue the embryonic lethality of homozygous mice with a hypomorphic $Brca1^{\Delta 11}$ allele (Hakem et al. 1997; Xu et al. 2001b). Moreover, these animals developed mammary tumors with loss of the remaining *Trp53* allele within 6–12 months (Xu et al. 2001b). The application of Cre-loxP technology to induce combined tissue-specific loss of *Brca1/2* and *Trp53* (and other tumor suppressors) has yielded more sophisticated mouse models of BRCA1/2-associated breast cancer (Xu et al. 1999; Jonkers et al. 2001; Liu et al. 2007a; McCarthy et al. 2007) and ovarian cancer (Szabova et al. 2012; Perets et al. 2013; Zhai et al. 2017). These models have been used to study the cell of origin (McCarthy et al. 2007), tumor–immune interactions (Ding et al. 2018; Pantelidou et al. 2019; Zimmerli et al. 2022), and driver genes in BRCA1/2-associated cancer (Annunziato et al. 2019), as well as for prevention studies (van de Ven et al. 2017) and interventions to study therapy response and resistance (Rottenberg et al. 2007, 2008).

Additionally, several *Brca1* mutant mouse strains have been generated to assess how specific aspects of BRCA1/2 biology contribute to observed phenotypes in vivo. For instance, a C61G mutation in the E3 ubiquitin ligase domain of BRCA1, which abolishes E3 activity and BRCA1–BARD1 interaction, caused embryonic lethality and loss of tumor suppression (Drost et al. 2011). In contrast, the I26A mutation, which reduces E3 activity but still allows formation of the BRCA1–BARD1 heterodimer, only caused male sterility and did not predispose animals to tumors (Reid et al. 2008; Shakya et al. 2011; Wang et al. 2023). These models provide insights into the importance

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of BRCA1–BARD1 heterodimer stability in tumor suppression. Similarly, a S1598F mutation in the BRCA1 BRCT domain (corresponding to S1655F in humans) that abolishes interaction with phosphorylated proteins such as CtIP, BRIP1, and Abraxas caused both male sterility and tumor acceleration in breast and pancreatic cancer models, highlighting the essential role of BRCT phospho-recognition in BRCA1 tumor suppression (Shakya et al. 2011). The L1363P mutation (L1407P in humans) disrupts the BRCA1 CC domain, important for interacting with PALB2 and BRCA2-mediated RAD51 loading, and caused embryonic lethality or an FA-like phenotype, depending on genetic background, as well as HR deficiency and increased cancer predisposition (Park et al. 2020; Pulver et al. 2021). Interestingly, compound heterozygous mice carrying one *Brca1*$^{\Delta 11}$ allele and one *Brca1*$^{\Delta CC}$ coiled-coil deletion allele are viable and display partial HR restoration, suggesting a separation of function between the role of BRCA1 in DNA end resection and its role in RAD51 recruitment (Nacson et al. 2020).

For BRCA2, mutant mice carrying a S3214A (SA) missense mutation in the C-terminal RAD51 interaction domain have been generated. This mutation disrupts the role of BRCA2 in stalled fork protection and ssDNA gap suppression but does not affect its role in HR. Importantly, homozygous *Brca2*$^{SA/SA}$ mice did not display genomic instability or accelerated tumor formation, showing that these phenotypes are driven by loss of HR and not by loss of fork protection or gap suppression (Lim et al. 2024).

### *BRCA1/2 haploinsufficiency*

Hereditary cancer susceptibility due to heterozygous *BRCA1/2* mutations can be explained by the Knudson two-hit model, in which individuals initially carry a monoallelic germline mutation and later lose the second allele somatically (Knudson 1971; Neuhausen and Marshall 1994; Collins et al. 1995). This process, known as loss of heterozygosity (LOH), may occur through various mechanisms, such as allelic recombination, genetic rearrangements, deletions, or promoter hypermethylation (Neuhausen and Marshall 1994; Moynahan and Jasin 1997; Osorio et al. 2002). In *BRCA1/2* mutation carriers, sporadic loss of the wild-type allele may induce genome instability that promotes the accumulation of additional cancer driver mutations. However, homozygous loss of these genes is lethal, suggesting that suppressor mutations are required first to provide a permissive genetic context for cells to undergo *BRCA1/2* LOH and neoplastic transformation.

The first observation that heterozygous germline mutations in HR genes cause abnormal DNA damage signaling and replication stress was made in cells from *PALB2* mutation carriers (Nikkilä et al. 2013). It is currently hypothesized that heterozygous *BRCA1/2* mutations also induce low levels of genomic instability, increasing the likelihood of acquiring additional mutations in other driver genes prior to *BRCA1/2* LOH. Indeed, analysis of primary cells from *BRCA1* mutation carriers revealed that *BRCA1* haploinsufficiency induces replication stress, genomic instability, and telomere shortening (Rennstam et al. 2010; Pathania et al. 2014; Sedic et al. 2015). Similarly, *Brca2* haploinsufficiency causes both defective replication fork protection and postreplicative ssDNA gap formation in mouse embryonic fibroblasts (Lim et al. 2024).

Haploinsufficiency-related genome instability and LOH-associated cell lethality might also reduce the selective pressure for LOH during *BRCA1/2*-associated tumorigenesis. In line with this, retention of the wild-type allele is observed in a considerable fraction of *BRCA2*-associated breast and ovarian cancers (16% and 46%, respectively) as well as in a small fraction of *BRCA1*-associated tumors (Maxwell et al. 2017). However, due to the small number of cases, it cannot be excluded that BRCA1-associated tumors retaining a wild-type copy may simply be sporadic tumors rather than tumors driven by *BRCA1* haploinsufficiency.

A preclinical study in mice showed that germline *Brca2* heterozygosity may promote the development of *Kras*$^{G12D}$-driven pancreatic tumors that retain the wild-type *Brca2* allele (Skoulidis et al. 2010). In patients, the absence of *BRCA1/2* LOH is associated with lower HR deficiency scores and reduced response to platinum chemotherapy, implying that LOH should be considered when selecting the most effective treatment option (Maxwell et al. 2017). Furthermore, it was recently reported that accumulation of the metabolite methylglyoxal may promote proteasomal degradation of BRCA2, suggesting that there may also be nongenetic mechanisms inducing (temporal) BRCA1/2 deficiency (Kong et al. 2024). A study in *BRCA2* mutation carriers revealed that noncancerous mammary tissues already displayed DNA copy number aberrations and levels of DNA damage, supporting the notion that *BRCA2* heterozygosity may foster genome instability (Karaayvaz-Yildirim et al. 2020).

*BRCA1* haploinsufficiency may be masked by RNF168, as *Brca1* heterozygous mice showed reduced viability and cancer predisposition on a *Rnf168* knockout background (Zong et al. 2019). This was explained by the ability of RNF168 to substitute for BRCA1 in recruiting the PALB2 and BRCA2 mediator complex to damaged DNA (Zong et al. 2019). Conceivably, conditional haploinsufficiency might be induced in certain tissues where RNF168 levels are low or *BRCA2* heterozygous cells are susceptible to aldehyde-induced proteasomal degradation of BRCA2 (Tan et al. 2017). Interestingly, single-cell analysis of mastectomies from *BRCA1/2* mutation carriers revealed a possible exhaustion of tissue-resident immune cells, suggesting that immune escape mechanisms might arise in noncancerous tissues during (or even prior to) tumor initiation (Reed et al. 2024). This is in line with lower counts of $CD8^+$ T lymphocytes detected in *BRCA1* mutation carriers compared with age-matched noncarriers (Wu et al. 2023); however, these findings require further validation in larger clinical cohorts.

Overall, the nuances of *BRCA1/2* haploinsufficiency are still poorly understood, and their contributions to *BRCA1/2*-associated tumorigenesis remain uncertain.

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Hence, more research is needed to provide critical insights.

### *Roles of BRCA1 and BRCA2 in meiosis*

Although the molecular functions of BRCA1/2 are extensively studied in somatic cells, their roles in meiosis have received far less attention. Early clues indicating that both factors are important for meiosis came from immunofluorescence studies in spermatocytes as well as from mouse models carrying hypomorphic *Brca1/2* alleles (Scully et al. 1997a,b).

Mice carrying $Brca1^{\Delta 11}$ alleles on a *Trp53* heterozygous background ($Brca1^{\Delta 11/\Delta 11}$;$Trp53^{+/-}$) are viable but show male infertility due to an arrest of spermatogenesis at the pachytene stage, accompanied by increased apoptosis (Xu et al. 2003). BRCA1 was subsequently shown to directly localize to unsynapsed chromosomes and mediate meiotic silencing of unsynapsed chromatin (MSUC) (Turner et al. 2004a; Kouznetsova et al. 2009). Importantly, studies using the hypomorphic $Brca1^{\Delta 11}$ allele suggested that BRCA1 is dispensable for meiotic recombination, as female $Brca1^{\Delta 11}$ mice are fertile (Broering et al. 2014). However, a recent study with viable *Brca1*-null mice carrying $Brca1^{\Delta 5-13}$ alleles on a *Trp53bp1* knockout background showed that BRCA1 is also essential for female fertility and euploidy of oocytes (Bai et al. 2024). Further experiments showed that BRCA1 promotes the recruitment of meiotic recombinases RAD51 and DMC1 to DSBs during meiotic recombination in both oocytes and spermatocytes as well as the elimination of recombination-defective oocytes via activation of the asynapsis checkpoint (Bai et al. 2024).

Due to the essentiality of BRCA2, several studies have focused on studying orthologs in animals and plants to reveal its function in meiosis. An early study reported infertility in *Brca2*-null mice expressing human BRCA2 due to low *BRCA2* expression in the gonads and demonstrated that BRCA2-deficient spermatocytes do not progress beyond early prophase I and display reduced RAD51 and DMC1 focus formation due to defective meiotic recombination (Sharan et al. 2004). Other studies showed that genetic loss or RNAi-mediated silencing of the *BRCA2* orthologs in *Arabidopsis thaliana*, *Drosophila melanogaster*, or *Caenorhabditis elegans* induced sterility resulting from disturbed HR in meiotic cells, indicating a key role of BRCA2 in meiosis (Siaud et al. 2004; Klovstad et al. 2008; Ko et al. 2008). Although PALB2 is responsible for the recruitment of BRCA2 to DSBs in somatic cells, a distinct meiotic localizer of BRCA2, MEILB2, has been identified in germ cells. Loss of MEILB2 in mice impairs the localization of BRCA2, RAD51, and DMC1 to ssDNA in spermatocytes, leading to male sterility (Zhang et al. 2019). Altogether, these studies establish an essential role for BRCA2 in meiotic HR.

### *Association with cGAS–STING pathway*

Loss of BRCA1/2 or other HR-related genes leads to defective HR and compromised genome maintenance, termed BRCAness (Turner et al. 2004b; Lord and Ashworth 2016). As described above, genome integrity in *BRCA1/2* mutant cancers is also affected by a defect in replication fork protection (Schlacher et al. 2011, 2012). In recent years, genomic instability has been associated with the activation of inflammatory signaling via the cyclic GMP–AMP synthase (cGAS) and stimulator of interferon genes (STING) pathway. Indeed, *BRCA1/2* mutations in ovarian cancer correlate with increased numbers of tumor-infiltrating lymphocytes, elevated STAT1 signaling, and higher expression of immune-related genes and immune checkpoint molecules PD1/PDL1 and STING (McAlpine et al. 2012; Strickland et al. 2016; Wieser et al. 2018; Bruand et al. 2021). A recent analysis of 380 primary breast cancer samples showed increased expression of cGAS–STING signaling components in TNBC samples compared with other subtypes. In addition, cGAS–STING expression scores correlated with HR deficiency, mutational burden, and high immune infiltration (Chen et al. 2024).

The cGAS–STING pathway senses microbial and host cell-derived DNA and RNA in the cytosol to trigger an innate immune response (Chen et al. 2016). BRCA1/2 loss leads to an increase in mitotic chromatin bridges and chromosome missegregation during mitosis, which may result in the formation and cytosolic accumulation of micronuclei that are recognized by cGAS, triggering STING inflammatory signaling (Harding et al. 2017; Mackenzie et al. 2017; Parkes et al. 2017). Indeed, it was shown that cGAS-positive micronuclei form upon BRCA2 loss and trigger a cGAS–STING response, enhancing sensitivity to tumor necrosis factor-α (TNFα) signaling (Heijink et al. 2019). Alternatively, a recent study suggests that aberrant R-loop processing may be a source of innate immune activation due to the accumulation of R-loops in the cytoplasm (Crossley et al. 2023). Therefore, the exact mechanism underlying innate immune activation in response to BRCA1/2 loss remains to be elucidated.

## Prevention and treatment of *BRCA1/2* mutated cancers

### *Prevention strategies for BRCA1/2 mutation carriers*

To reduce the elevated cancer risk of *BRCA1/2* mutation carriers, risk-reducing surgeries may be performed. The most effective prophylactic approach is risk-reducing mastectomy, which involves the removal of one or both breasts and lowers the risk of breast cancer in mutation carriers by up to 90% (Metcalfe et al. 2014). Similarly, risk-reducing salpingo-oophorectomy (RRSO), involving the removal of the ovaries and fallopian tubes, can reduce the risk of ovarian cancer by >96% (Gootzen et al. 2024). However, these surgeries have a massive impact on the quality of life, as RRSO induces "surgical menopause," leading to adverse effects such as increased fatigue, anxiety, impaired cardiovascular health, and insomnia (Rocca et al. 2009; Vermeulen et al. 2017; Stuursma et al. 2018). Therefore, hormone replacement therapies are often prescribed to mitigate these effects (Armstrong et al. 2004; Gordhandas et al. 2019).

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Alternative means exist to manage cancer risk in mutation carriers opting against surgeries, but they remain poorly studied. Besides regular screening using mammography and magnetic resonance imaging, selective estrogen receptor modulators or aromatase inhibitors have been suggested to reduce the risk of breast cancer, though further large-scale studies are needed to confirm this observation (Kotsopoulos et al. 2023). Some reports suggest that breastfeeding may reduce cancer risk in *BRCA1/2* mutation carriers, possibly because of an increased breast tissue differentiation and reduced estrogen exposure (Kotsopoulos et al. 2012). Oral contraceptives have also been suggested to reduce the risk of ovarian cancer due to their proapoptotic effect on ovary and tubal epithelial cells (Moorman et al. 2013; Xia and Kotsopoulos 2022). Furthermore, RANKL/RANK pathway inactivation has been shown to delay mammary tumor onset in animal models (Sigl et al. 2016), and biopsies from patients treated with the RANKL inhibitor denosumab showed reduced proliferation (Nolan et al. 2016). A currently ongoing clinical trial comparing the effect of denosumab versus placebo treatment will reveal whether this agent can be used as a preventive strategy for *BRCA1* mutation carriers (NCT04711109). Overall, the above-mentioned surgical and systemic strategies are important tools to enhance the quality of life and longevity for *BRCA1/2* mutation carriers; however, further research is needed to offer improved alternatives to highly invasive surgeries.

### *Therapeutic targeting of BRCA1/2 mutated cancers*

As mentioned above, BRCA1/2-deficient tumors are sensitive to ICL-inducing agents, such as platinum-based chemotherapies (Rubin et al. 1996; Bhattacharyya et al. 2000). Cisplatin and carboplatin generate monoadducts as well as DNA cross-links and were clinically approved as a first-line treatment for various cancer types already in 1978 (Kelland 2007). Initial studies in mouse models of BRCA1-deficient breast cancer showed that animals displayed a favorable response to cisplatin treatment (Rottenberg et al. 2007). These findings align with a recent clinical study in TNBC patients, which reported that carboplatin had double the objective response rate of docetaxel in *BRCA1/2* mutation carriers, whereas there was no difference in an unselected population (Tutt et al. 2018).

Despite the initial sensitivity of *BRCA1/2* mutated tumors to platinum drugs, their efficacy is hindered by dose-limiting toxicity in normal tissues and the frequent development of resistance (Martin et al. 2008; Rottenberg et al. 2021). It was demonstrated early on that BRCA2-deficient tumor cells acquire platinum resistance through secondary intragenic mutations in *BRCA2* that restore the wild-type reading frame, resulting in re-expression of functional BRCA2 protein (Edwards et al. 2008; Sakai et al. 2008). Similar frame-restoring somatic mutations in both *BRCA1* and *BRCA2* have been found in platinum-resistant *BRCA1/2* mutated ovarian cancers, highlighting the clinical relevance of this resistance mechanism (Swisher et al. 2008; Sakai et al. 2009; Norquist et al. 2011).

The primary goal of all treatment strategies for cancer is to target and eliminate tumor cells while sparing healthy tissues. This may be achieved by synthetic-lethal therapies based on targeted inhibitors that selectively kill tumor cells with a particular molecular defect. A prime example of synthetic lethality in cancer was observed by two landmark studies conducted 10 years after the discovery of BRCA1/2. These studies demonstrated that small-molecule inhibitors of poly(ADP-ribose) polymerase 1 (PARP1) selectively kill BRCA1/2-deficient cells (Bryant et al. 2005; Farmer et al. 2005). This breakthrough discovery spurred the development of multiple PARP inhibitors (PARPis), resulting in the clinical approval of six PARPis (olaparib, rucaparib, niraparib, and talazoparib in the United States and Europe, and fuzuloparib and pamiparib in China) for the treatment of patients with BRCA1/2-deficient tumors (Mateo et al. 2019; Zeng et al. 2024).

PARP1 is a nuclear enzyme essential for detecting single-stranded DNA breaks (SSBs) and DSBs. Upon binding to DNA lesions, PARP1 undergoes autoPARylation and facilitates the recruitment of downstream DDR proteins (Ray Chaudhuri and Nussenzweig 2017). Initially, it was believed that PARP1 loss results in an accumulation of SSBs, which during DNA replication results in the formation of DSBs that cannot be repaired in the absence of functional HR, leading to mitotic catastrophe and cell death (Bryant et al. 2005; Farmer et al. 2005). However, this hypothesis was soon reconsidered, as genetic depletion of PARP1 did not show an increase in SSBs. Additionally, the loss of XRCC1, a factor acting downstream from PARP1, was also synthetic lethal with PARP inhibition (Strom et al. 2011). This led to the hypothesis that PARPis trap PARP1 onto DNA, forming noncovalent DNA–protein cross-links that induce replication fork collapse and accumulation of DSBs. In the absence of HR, these breaks are then repaired by more mutagenic repair processes (Murai et al. 2012). This concept is reinforced by the observation that different PARPis exhibit varying cytotoxicity effects despite inhibiting PARP1 catalytic activity to a similar extent (Murai et al. 2014; Zandarashvili et al. 2020). However, this has been challenged by the observation that PARPi treatment causes ssDNA replication gaps, which have been suggested to be sufficient to induce toxicity in the absence of BRCA1/2 (Cong et al. 2021; Paes Dias et al. 2021). Alternatively, PARP inhibition has been proposed to induce transcription–replication conflicts, resulting in the accumulation of DNA damage in HR-deficient cells regardless of PARP trapping (Petropoulos et al. 2024).

All currently approved PARPis are first-generation inhibitors that are not PARP1-selective and essentially equipotent against PARP1 and PARP2. Consequently, the clinical utility of these drugs has been limited by hematological toxicities associated with the simultaneous inhibition of PARP1 and PARP2, prompting the need for more selective PARP1 inhibitors (Illuzzi et al. 2022; Liu et al. 2023). Recently, a second-generation PARP1-specific inhibitor and PARP1 DNA trapper, saruparib/AZD5305, was developed, which demonstrated strong efficacy in a *BRCA2* mutant patient-derived xenograft (PDX) breast

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cancer model (Johannes et al. 2021). Additional preclinical studies showed that saruparib is strongly PARP1-selective and efficacious as a monotherapy or in combination with carboplatin, suggesting that specific inhibition and trapping of PARP1 might reduce adverse effects while maintaining a high therapeutic efficacy (Illuzzi et al. 2022; Dellavedova et al. 2023; Herencia-Ropero et al. 2024).

### *PARP inhibitor resistance in BRCA1/2 mutated cancers*

PARPis have demonstrated strong clinical responses, leading to a significantly extended progression-free survival in multiple clinical trials (Geyer et al. 2022). However, most patients with advanced cancer ultimately develop resistance, likely fueled by the high genomic instability of *BRCA1/2* mutant tumors. Various mechanisms of acquired resistance have been identified and characterized using a combination of clinical data and preclinical models, though their prevalence in clinical practice remains to be fully determined (Fig. 4).

Initial studies with olaparib in mouse models developing BRCA1-deficient mammary tumors revealed drug extrusion via upregulation of the drug efflux transporter ABCB1/P–glycoprotein (P-gp) as one of the first mechanisms of PARPi resistance (Rottenberg et al. 2008; Jaspers et al. 2015). The clinically most described mechanism of PARPi resistance is the occurrence of *BRCA1/2* reversion mutations that restore the reading frame and lead to re-expression of a functional BRCA protein (Edwards et al. 2008; Sakai et al. 2008). Several studies have identified BRCA reversion mutations in clinical samples and have also linked them to cross-resistance to platinum agents; however, these findings remain to be validated in larger cohorts (Weigelt et al. 2017; Lin et al. 2019; Waks et al. 2020; Harvey-Jones et al. 2024). Also, the notion that the hypomorphic activity of certain *BRCA1* alleles can drive resistance to PARPis and/or platinum drugs still awaits clinical validation (Drost et al. 2011, 2016; Wang et al. 2016a,b).

Additionally, BRCA1-deficient cells may acquire PARPi resistance through HR restoration by loss of classical NHEJ-promoting factors, such as 53BP1 (Bouwman et al. 2010; Bunting et al. 2010). Later studies have identified that loss of factors acting downstream from 53BP1 (i.e., RIF1, REV7, and members of the shieldin complex) also re-establish DNA end resection in BRCA1-deficient cells (Di Virgilio et al. 2013; Escribano-Díaz et al. 2013; Feng et al. 2013; Xu et al. 2015; Dev et al. 2018; Ghezraoui et al. 2018; Gupta et al. 2018; Noordermeer et al. 2018). Besides 53BP1 and its downstream factors, loss of the CTC1–STN1–TEN1 (CST) complex also causes PARPi resistance in BRCA1-deficient cells, albeit to a milder extent than loss of 53BP1 and Shieldin complex members (Barazas et al. 2018; Mirman et al. 2018). Other factors for which loss of function has been proposed to cause PARPi resistance in BRCA1-deficient cells via restoration of DNA end resection are HELB, DYNLL1, and ERCC6L2; however, their exact mechanism of action remains

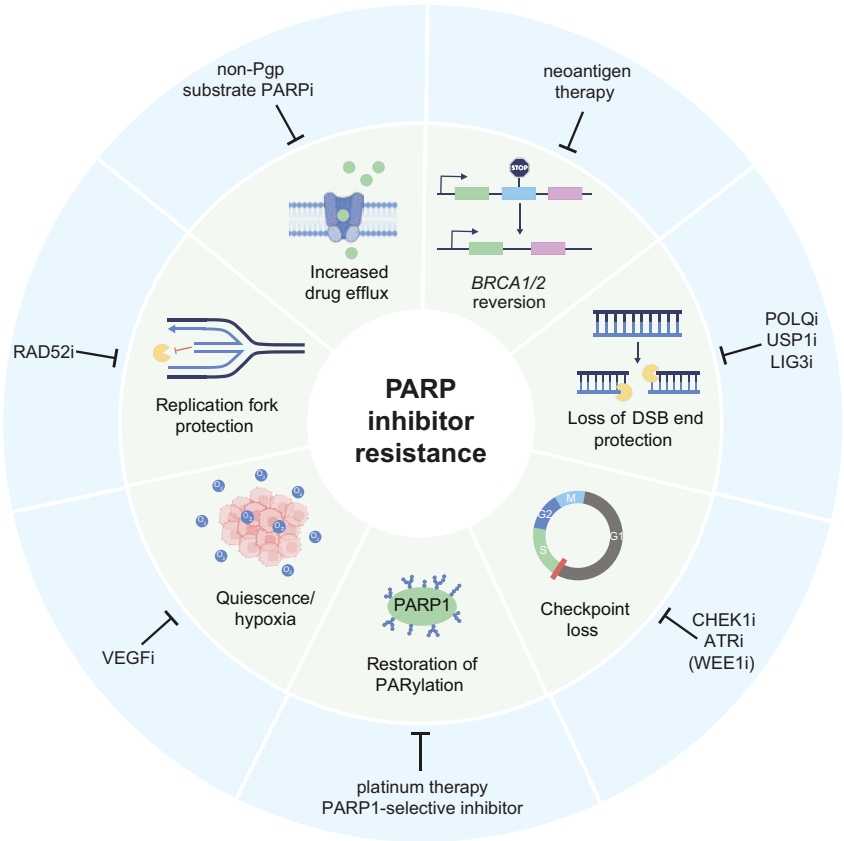



**Figure 4.** Mechanisms of acquired resistance to PARPis and strategies to overcome it. The inner circle represents various mechanisms of PARPi resistance, whereas the outer circle depicts possible strategies to overcome PARPi resistance. There are several mechanisms of PARPi resistance that can be categorized as follows: upregulation of drug efflux pumps, BRCA1/2 reversion mutations, loss of DSB end protection, loss of cell cycle checkpoint arrest, restoration of PARylation, entering states of hypoxia/quiescence, and restoration of replication fork protection.

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unclear (Tkáč et al. 2016; Becker et al. 2018; He et al. 2018; Francica et al. 2020).

As mentioned above, loss of BRCA1/2 causes degradation and collapse of stalled replication forks. It is therefore not surprising that restoration of replication fork stability has been proposed as a PARPi resistance mechanism. Loss of SNF2 family fork remodelers such as SMARCAL1 may reduce PARPi sensitivity by reducing reversal and MRE11-mediated degradation of stalled replication forks in BRCA1/2-deficient cells, but it is unclear whether this can drive PARPi resistance in a clinically relevant setting (Taglialatela et al. 2017).

Finally, modifications of PARP1 itself may also reduce PARPi efficacy. CRISPR screens revealed that mutations in PARP1 cause PARPi resistance by altering PARP1 trapping efficacy (Pettitt et al. 2018). PARP1 phosphorylation at Tyr907 by MET may enhance PARP1 enzymatic activity, rendering cells resistant to PARPis (Du et al. 2016). Similarly, loss of the PAR glycohydrolase PARG may restore PARylation in PARPi-treated cells, leading to PARPi resistance (Gogola et al. 2018).

Although the use of PARPis has led to significant advances in the clinical management of BRCA1/2-deficient tumors, the emergence of drug resistance poses a major challenge to effective treatment. To increase the efficacy of PARPis and extend the benefits of these drugs to more patients, it will be crucial to analyze data from ongoing trials, identify and understand the clinically relevant resistance mechanisms, and design more effective combination therapies while managing toxicity.

### *Combating therapy resistance in BRCA1/2 mutant cancers*

As the use of PARPis is expanding and moving to earlier stages of therapy, strategies to increase efficacy and overcome resistance have become an important area of research. Several studies have explored strategies to combat resistance, and numerous clinical trials are currently evaluating combination therapies, which may increase treatment opportunities for patients with *BRCA1/2* mutant tumors (Fig. 4).

Inhibition of vascular angiogenic growth factor (VEGF) has been associated with downregulation of BRCA1/2 and RAD51 in ovarian cancer cell lines (Lim et al. 2014; Kaplan et al. 2019). Interestingly, a combination of the VEGF inhibitor bevacizumab and PARPis has already demonstrated clinical efficacy in ovarian cancer and may be a rational approach to increase efficacy and overcome PARPi resistance (Mirza et al. 2019; Hardesty et al. 2022; Ray-Coquard et al. 2023).

Acquisition of PARPi resistance through the restoration of HR or rescued replication fork stability is often partial and generally induces novel vulnerabilities that may be targetable. One of these targets may be the ATR kinase, which has been shown to be activated upon PARPi exposure. ATR is the major kinase activated upon replication stress and activates many downstream targets such as CHK1 to induce checkpoint arrest and repair of stalled forks (Ngoi et al. 2021). Promising preclinical data led to the development of ATR inhibitors that are currently being evaluated in several clinical trials (NCT03462342 and NCT04972110) (Kim et al. 2017, 2020; Yazinski et al. 2017).

Exploring alternative synthetic-lethal interactions in *BRCA1/2* mutant tumors offers a viable strategy to selectively target BRCA1/2-deficient tumors and enhance the efficacy of PARPis (Fig. 5). BRCA1/2-deficient cells exhibit heightened sensitivity to genomic insults that disrupt replication fork progression, such as PARP inhibition; however, other types of lesions may serve as a similar barrier to faithful replication. One such example is base

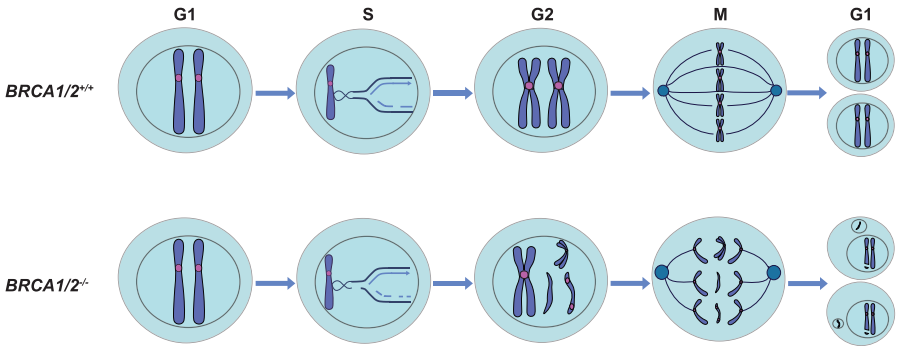



**Figure 5.** Synthetic-lethal interactions in BRCA1/2-deficient cells. (*Top*) A BRCA1/2-proficient cell that is proficient in protection of stalled replication forks and capable of repairing DSBs occurring during S phase via HR, resulting in the generation of two daughter cells with intact sets of chromosomes. (*Bottom*) A BRCA1/2-deficient cell that lacks fork protection and HR. This cell is vulnerable to replication stress and relies on error-prone pathways for DSB repair during S phase, resulting in persistent DNA lesions that induce mitotic aberrations (lagging chromosomes, chromatin bridges, or ultrafine DNA bridges) and daughter cells with missegregated chromosomes. Lagging chromosomes or chromosomal fragments can end up in micronuclei, where they may undergo chromothripsis and complex intrachromosomal rearrangements. Synthetic lethality in BRCA1/2-deficient cells can be induced via several mechanisms. First, the induction of replication-blocking lesions, such as base damage or ssDNA gaps, may cause replication fork stalling and the accumulation of DSBs. Second, error-prone MMEJ may be targeted to prevent the repair of DSBs. Third, mitotic tethering of chromosome fragments can be inhibited. All these synthetic-lethal mechanisms cause increased genomic instability and micronucleus formation, thereby triggering death via mitotic catastrophe in BRCA1/2-deficient cells.

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damage remaining unrepaired in the absence of the chromatin remodeler ALC1 or the APE2 nuclease. Additionally, the accumulation of genomic ribonucleotides upon RNASEH2 depletion may induce genomic instability in the absence of BRCA1/2 (Zimmermann et al. 2018; Mengwasser et al. 2019; Álvarez-Quilón et al. 2020; Hewitt et al. 2021; Verma et al. 2021). Similarly, the accumulation of ssDNA gaps in the absence of DNA ligase III or USP1 may also be detrimental in BRCA1/2-deficient cells, and several USP1 inhibitors are currently in clinical development (Lim et al. 2018; Paes Dias et al. 2021; Da Costa et al. 2024). Depletion of FANCM, which mediates the repair of stalled replication forks, has also been reported to be synthetic lethal with BRCA1/2 loss, though the underlying mechanism remains unclear (Pan et al. 2017; Panday et al. 2021).

Another vulnerability of HR-deficient cells is their increased reliance on alternative DSB repair pathways, such as the error-prone MMEJ pathway, which is dependent on polymerase θ (POLQ) (Chan et al. 2010; Ceccaldi et al. 2015; Mateos-Gomez et al. 2015). Inhibition of POLQ presents an attractive strategy to target BRCA1/2-deficient cells and has also demonstrated toxicity in PARPi-resistant cells with restored HR due to loss of 53BP1/Shieldin. Thus, POLQis may also synergize with PARPis. Several approaches are being explored to specifically inhibit either the polymerase domain (ART558) or the helicase domain (NVB) of POLQ (Zatreanu et al. 2021; Zhou et al. 2021). POLQis have also been reported to induce the accumulation of ssDNA due to increased DNA end resection, consequently leading to dysfunctional repair (Belan et al. 2022; Schrempf et al. 2022). Several clinical trials are currently under way to assess the efficacy and tolerability of POLQis alone or in combination with PARPis.

The reliance of BRCA1/2-deficient cells on error-prone DSB repair pathways also results in elevated levels of fragmented chromosomes, which leads to chromosome missegregation during mitosis, resulting in genomic rearrangements and micronucleus formation (Feng and Jasin 2017). These segregation defects appear to be largely mitigated by the mitotic tethering of fragmented chromosomes, ensuring their segregation to the same daughter cell (De Marco Zompit et al. 2022; Trivedi et al. 2023). Disruption of the CIP2A–TOPBP1 complex, essential for mitotic tethering, has been reported to induce synthetic lethality in *BRCA1/2* mutated cells by promoting lethal missegregation of acentric chromosomes (Adam et al. 2021; Laine et al. 2021).

## Concluding remarks

In summary, BRCA1 and BRCA2 play multifaceted roles beyond their well-established functions in HR. They are essential for maintaining genomic stability under various conditions, such as replication stress, ICLs, and during meiosis and mitosis. Despite intense research efforts, several open questions remain about how BRCA1/2 act precisely during these diverse cellular processes and which of these mechanisms ultimately contribute to tumorigenesis.

Although it has been demonstrated recently that *BRCA1/2* LOH is only detected in tissues known to be affected in *BRCA1/2* mutation carriers, such as breast and ovarian tissues (Jonsson et al. 2019), the reasons for the tissue tropism of *BRCA1/2* mutant cancers remains a conundrum. Several hypotheses have been proposed since the early 2000s; experimental progress validating these ideas has been limited (Elledge and Amon 2002). One theory suggests that genetic redundancy occurs in other tissues, masking the consequences of BRCA1/2 deficiency; however, this seems unlikely considering the multifunctionality of BRCA1/2 in maintaining genome stability. Alternatively, breast and ovarian tissues may be unique in tolerating BRCA1/2 loss due to additional genetic or environmental factors such as hormonal regulation. For example, estrogen signaling may promote the survival of BRCA1/2-deficient cells (Gorrini et al. 2014) or induce additional DNA damage that may elevate genomic instability and tumorigenesis (Savage et al. 2014). Furthermore, the role of *BRCA1/2* haploinsufficiency in tissue specificity remains poorly understood. Additional research is urgently needed to better understand the tissue-specific nature and the precancerous stages of *BRCA1/2*-associated tumorigenesis.

Currently, prevention strategies for *BRCA1/2* mutation carriers include preventive mastectomy and prophylactic oophorectomy, which are highly invasive procedures for patients. This underscores the urgent need for the development of alternative, noninvasive prevention options. Additionally, despite their strong (pre)clinical activity, PARPis and platinum agents are unable to fully eradicate *BRCA1/2* mutant tumors in advanced disease settings. Durable complete remissions have only been achieved with high-dose alkylators, which are associated with significant toxicity (Huang et al. 2009; Evers et al. 2010; Osher et al. 2011; Vliek et al. 2023). It will therefore be important to explore alternative treatment options for patients with BRCA1/2-deficient tumors. Next-generation PARPis and novel therapeutic strategies to inhibit TOPBP1–CIP2A, POLQ, USP1, and LIG3 may exploit vulnerabilities of BRCA1/2-deficient cells while sparing healthy tissues. These improved PARPis or novel synthetic-lethal therapeutics may be used as single agents or in combination with existing therapies, provided that side effects and toxicity remain manageable. Furthermore, functional and clinical studies have shown that *PALB2* mutant cells and tumors display phenotypes similar to those observed for *BRCA2*-deficient cells. Therefore, it is anticipated that treatment and preventive strategies used for *BRCA2* mutation carriers may also be effective for individuals with *PALB2* mutations.

Ideally, targeted therapies should suppress rather than enhance genomic instability of BRCA1/2-deficient tumors to prevent acquisition of mutations that cause resistance. Continued research is needed to refine and improve the efficacy of existing treatments and develop novel therapeutic strategies for improved clinical management of patients with *BRCA1/2* mutant tumors.

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## Competing interest statement

The authors declare no competing interests.

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